4. Design of the Job

The Worker's Function in the Workplace

The major question that should be asked is: "What does the worker do on the job?" If the job can be broken down into functions, the activities and tasks that are performed can be specified. These activities and tasks can then be analyzed. The analysis of the tasks provides information as to the optimum job design.

There are many purposes for the design or redesign of a job. The purpose of the design or redesign depends on the objectives that we wish to obtain. Among the possible objectives for job design are:

1. Increased efficiency
2. Increased productivity
3. Increased worker satisfaction
4. Decreased potential for hazard exposure

Though each of the first three reasons for job design is important in its own right, only the fourth reason for job design will be discussed here. Thus, the major objective of job design as related to this discussion is to decrease the potential hazard exposure of the worker on the job.

The Job Functions. Now let's look further to determine the answer to our original question; that is, "What does the worker do on the job?" Perhaps in determining the job functions that the worker performs, general rules for controlling the worker's exposure to potential hazards can be described.

The first step that the worker performs is to gather data and information on the situation as it currently exists. This data-gathering function may be limited, as is the case when a worker is performing basically a physical task and must only observe such things as obstacles in his path, the placement of hand tools, etc. On the other hand, the airline pilot is faced with a multitude of data and information that must be presented in a reasonable manner to allow for appropriate pilot action to control the airplane.

The second step that the worker must perform is the processing of the data and information that have been gathered. The purpose of this data processing is to make decisions concerning the future steps that will be taken. Again, in many cases this is a relatively simple process. Quite often, as is the case when driving an automobile, although basic data processing requirements may be somewhat complex, the actual processing itself goes on in the

background as an almost conditioned response to certain stimuli that the driver has learned to process through experience. On the other hand, in cases such as nonstandard or emergency situations, data processing can become quite complex and the determination of the appropriate response to the system status can be critical to the health and safety of the individual as well as to others in the vicinity.

The next major step in performing a job is to implement the decision that has been reached in the data processing stage. This decision can be implemented in two separate ways. First, the decision can be communicated to another worker for final action. A second method is for the worker who makes the decision to take the action to control the situation.

The three functions discussed above are basically cognitive processes; that is, they involve the worker's mental process. In many jobs it is also necessary for the worker to perform physical tasks. These tasks may be required to implement the decision or they may be of a supportive nature. Typical of the types of physical tasks that might be performed are the physical operation of controls to maintain or adjust the system status, the physical movement and positioning of the materials used, and the application of various types of tools to the object in order to attain the desired result.

Finally, while performing the functions of the job, it may be necessary for the worker to assume different positions in the workplace. In some cases, the worker may maintain a stationary position with only limited movement. Other types of jobs require the worker to be active with considerable movement throughout the work area.

A General Procedure for Determining Where Controls for Hazard Exposures Are Required

The first step in examining a job to determine where controls for hazard exposure are necessary is to review the functions and list the tasks that arerequired. As mentioned previously, it may be necessary to break the tasks further into elements and micromotions depending on the depth of the analysis needed.

Once the tasks have been listed and broken down into their components as necessary, the tasks or components should be analyzed to determine the requirements involved. What are the mental requirements that are present within the system? What information is required? What decisions must be made by the worker? What are the physical requirements of the jobs in terms of operating controls and other physical worker actions?

Now each of the individual tasks or components of the tasks should be analyzed to determine if potential hazard exposures exist. Are the requirements such that the worker cannot process the information that is presented? Is the worker provided with adequate information? Do the physical requirements of the job put an excess stress on the worker?

Next, alternate modes for performing the task, such as equipment, procedures, etc., should be identified where potential hazards exist. Can

substitution of method be used to remove the potential hazard? Can the worker be removed from the job totally, thus eliminating any potential hazard? Can new equipment or materials be provided that are less likely to involve a hazard to the worker? Each individual task or component of a task in which a potential hazard is identified should be analyzed to determine the changes that might be made.

Finally, before selecting the appropriate alternatives to use, each of the alternatives presented should be compared in terms of the costs and benefits of the alternative. In addition, any potential hazards that may exist in the alternative proposed for substitution should also be considered. The substitution of one hazard for another is not good design. Finally, the efficiency of the job itself should be determined using the alternative methods. Even though a particular method may lead to a lower hazard potential, it will be difficult to sell if efficiency is adversely affected.

Now let us look at each of the major functions of the job in more detail to determine how the job might be redesigned to remove potential hazards.

The Data Gathering Function

One of the major objectives of the design of jobs is to provide the worker with the data that are required for efficient functioning on the job. In orderto do this, one must ask the question, "What data are required?" What does the worker need to know to perform the job? Without adequate data, it is virtually impossible to perform a job efficiently. One cannot drive an automobile without having a view of the road through the windshield. Consider what would happen if the hood latch broke and the hood flew up to obscure your view while you are driving an automobile. You no longer have sufficient data to make a decision concerning the course upon which you are steering the automobile. Unless quick action is taken to remove the automobile from the stream of traffic and stop, there is a high probability that an accident will occur. Likewise, without adequate data, the worker is operating blindly in performing the job. It is highly probable that something will go wrong as a result of this lack of data.

After determining what data are required, the next question that should be asked is, "In what form should the data be provided to the worker?" As previously discussed, the form of the data may be qualitative or quantitative. The data may be presented as a deviation from normal or as a warning when action is required. An analysis of the method of presentation of data is important. Qualitative data are useful in some situations. However, in some cases it is necessary that exact values and the indication of trends be presented to the worker so that action can be taken before it is too late. In these situations, quantitative data are necessary. Each of the various types of data that are to be presented to the worker should be analyzed to determine the form of the presentation. Care should be taken to provide only those data that are necessary for performance of the job and to assure that the data that are presented, are presented in the appropriate mode needed for operator decision.

The next step is to determine when the data are required. Is it necessary that the operator receive the data on a continuous basis, or can an

intermittent supply of data be utilized? Again, using the example of driving an automobile, it is desirable that the operator have a clear view of the road ahead. However, intermittently the operator should view the rear of the automobile through the rear view mirror to assure that any traffic that is approaching from the rear is observed. In this case, some data are required continuously and some data are required intermittently.

The job designer should consider how much data are being presented to a given worker at a given point. Too much data can lead to a possible stressfulsituation in the worker. The visual and auditory channels can become overloaded, causing the worker to fail as a controller of the system. In such situations, it may be possible to modify the type of data presentation from a quantitative to a qualitative mode. Warning lights and auditory warning signals might be used to direct the worker's attention to a particular visual display, thus limiting the need for continuous monitoring of all displays. Finally, it may be necessary that additional workers be used to monitor the displays that are present.

Next, the job designer should consider how the data should be presented. Can the worker obtain the necessary data from the environment, or is a display necessary? If a display is required, should it be auditory or visual? What type of display should be used? Should a visual gauge be used? Consider the various kinds of displays available. Should the display be circular, semi-circular, vertical, horizontal, or digital? Consider the location of the display within the operator's viewing area. If a number of common displays are required, how can these be grouped in a manner to limit the necessity for continuous monitoring of each individual display?

Next, the need for an emergency warning system should be investigated. Should such a warning system be auditory or visual? At what point should such a warning system be provided in order to allow time for appropriate operator action to bring the system back into control?

In an emergency situation when a warning has been given to the operator, the designer must consider whether the operator can handle the abnormal conditions. What must the operator perform in an emergency situation? Can the operator perform this within the time allowed? If not, it may be necessary to provide a warning to the operator sooner, even though in some cases such a warning may prove to be a false alarm.

There are various ways in which the data-gathering phase can be modified to eliminate potential hazards or accidents. One way is to replace the human worker with automatic sensors. These sensors can provide for automatic shutdown when the situation so merits. Complete automation of the process, removing the need for worker interaction with the system, can be accomplished in such a manner. This is particularly useful in an emergency situation when it is impossible for the operator to react quickly enough to perform the many requirements to bring the system back under control.

The second method of modification can be to remove the monitoring of the process to a single location remote from the process itself. In this manner, an individual can monitor a number of systems at the same time. Again,

however, it is necessary to assure that this individual does not become overloaded.

When one of the senses becomes overloaded, it may be necessary to provide displays using one of the other senses. Most commonly, the visual sense will become overloaded; and in such situations, the use of auditory signals, vibration, or touch may be helpful to provide the worker with information necessary to control the system.

Replacement of quantitative scales with qualitative displays or go-no-go type displays may be advantageous. In this manner, the worker gathers information in a simple form indicating when situations arise requiring operator action.

The various types of data-display techniques should be analyzed to determine the display facilities that are best for the particular situation. Although in many cases the circular and semi-circular gauge are preferable for a quantitative display, this is in no way a hard and fast rule. Vertical and horizontal displays may provide the worker with the information more quickly. The digital display, which has the advantage of accuracy of reading, can be used in certain situations.

Finally, the designer must consider the need for redundancy. Should a breakdown occur in one of the information gathering channels, what backup is available to the worker to sense that something has gone wrong? What backup can be implemented if the worker makes an error? What is the potential loss that will be incurred if an error occurs and the system goes out of control? Redundancy can be provided to take care of these situations. The redundancy that is provided may be either parallel redundancy or series redundancy.

Processing Information

Once the data have been received by the worker, these data must be processed. Of concern to the designer is the time that is necessary to process the required data. Is sufficient time available to meet the time requirements for data processing? How much of this time is reaction time, and how much of this time is processing time? Can either the reaction time or the processing time be reduced by changes in the system? Finally, are the decisions that are required of the operator reasonable, considering the human information processing capabilities of the operator?

In this modern era of electronic data processing, the job designer should consider whether the data to be processed is best done by a human operator or by a machine or computer. There are certain general principles that will help the designer to make the appropriate determination.

In general, machines are best to perform the following functions.

1. Machines are best to sense low-level stimuli or small changes in the environment for certain types of stimuli.
2. Machines are best to handle standard situations without errors.
3. Machines are best to store and retrieve information quickly.

4. Machines are best to process large quantities of data.
5. Machines are best to make rapid and accurate calculations repetitively.
6. Machines are best to perform multiple functions simultaneously.
7. Machines are best to exert large physical forces. (Though not related to processing of data, this fact and should be considered.)
8. Machines are best to withstand stress without a decrement in service, if an appropriate maintenance level is provided.
9. Machines are best to operate continuously without fatigue.
10. Machines are best to operate in a patterned manner, given known input and desired outputs.

On the other hand, humans provide certain advantages in the workplace, particularly in terms of data processing. In general--

1. Humans are best to sense changes related to visual, auditory, tactual, olfactory, and taste senses.
2. Humans are best to handle the unexpected.
3. Humans are best to store and recall principles and decision rules.
4. Humans are best to adapt procedures to meet the situation.
5. Humans are best to put contingency plans into effect.
6. Humans are best to perform various tasks which are required randomly.
7. Humans are best to apply new methods to create new solutions.
8. Humans are best to develop priorities for action.

In general, if the problem is a structured, repetitive one, a machine will be best in handling it. On the other hand, if the problem is unstructured and variable, the human operator is the best.

Controlling the System

The next major function of the worker in the workplace is to control the system. The first question the job designer must ask is, "What kind of controls are required?" This is dependent on the type of adjustment that must be made to the system by the operator. Some adjustments require intermittent control action by the operator; in other situations, it is necessary that a continuous adjustment to the system be made. Thus, the type of control that is required may be either in discrete settings or continuous settings, depending on the type of control action that is necessary.

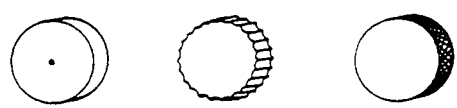
The designer must decide where to place the control and identify it so that there is little chance for error on the part of the operator selecting the control that is used. Can the operator look for the control? If so, labeling of the controls or coding them may be sufficient to provide a key as to the type of control that the operator is using. If the operator must find the controls without looking, it will be necessary to code the controls in some manner. In such a situation, it is possible to code the controls by location. The controls should be spaced apart using standard location patterns where possible. Another method of coding the controls is to code by shape. The shape may be symbolic of the object being controlled as is often the case of airplane controls. In any case, the shapes should be

significantly different from one another to assure that errors are not made. Where shape coding is used, it is advisable to conduct tests to determine those designs that are least likely to be confused.

Another method of coding controls is by size. Generally, a one-half difference in diameter and three-eighths difference in thickness will be sufficient to allow for discrimination between different controls. Finally, the coding of controls by texture has been used with some success. Research has indicated that three general types of texture coding can be discriminated by operators. These are a smooth surface, a fluted surface, and a knurled surface.

Figure 7.4.1

Texture coding.



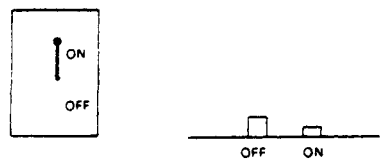
If the mode of operation is discrete, there are two basic kinds of controls that might be used. First, if the situation is dichotomous, an on-off control can be used. Such a control can be a button that when in the down position is "on" and in the up position is "off." An additional visual cue might be provided by lighting behind the button when it is in the "on" position. A simple switch is another alternative for the on-off situation. In such a case, when the switch is in the up position it is on and may be labeled and lighted to provide an additional visual cue.

Where multiple discrete settings are required, a rotary selector switch with an arrow pointing to a marker for the appropriate setting can be implemented. Another example is the use of a lever that can be positioned to obtain various discrete settings. In either case, to provide for protection against inadvertent setting to the wrong position, the positioning may require two motions. Such a situation exists in the manual gear-shift lever of an automobile.

If the mode of operation is continuous, various types of control mechanisms can be used. The round knob can be used where it is desirable that the control be operated by the fingertips or hand. A hand wheel can be used where a larger force is required, and arm motion is necessary to provide the appropriate system response. In some cases, a crank may be necessary to control the system response. Where the hands are busy or wnere large forces are required, a foot pedal can be used to obtain the necessary control. Finally, a lever can be used as a method for control of the system.

Figure 7.4.2

Discrete controls--on-off.



DISCRETE CONTROLS
MULTIPLE SETTINGS

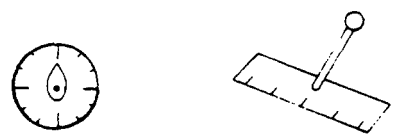


Figure 7.4.3

Continuous controls.



The Recommended Direction of Control Movements. There are recommended directions for control movements based upon previous conditioning that has occurred in humans. In general, humans expect controls to operate in certain manners. These recommendations should be implemented in the design of controls for the system.

In the case of on-off controls, the following directions as presented in Figure 7.4.4 are recommended.

General Control Design Principles

The following general control principles have been adapted from the book, Human Engineering Guide to Equipment Design, VanCott and Kinkade, 1972.

Figure 7.4.4

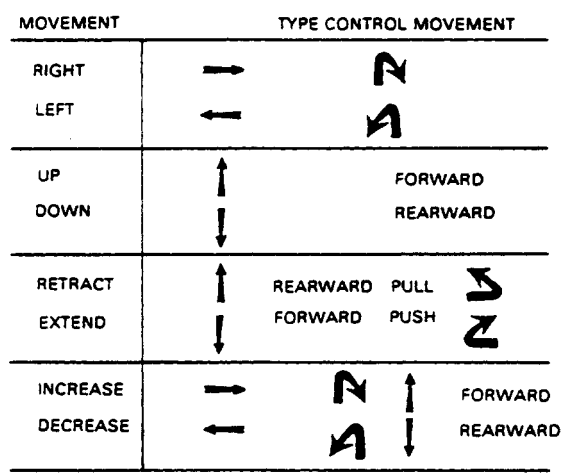
On-off control movement.

ON	OFF
Up	Down
In	Out
Right	Left
Pull	Push

For continuous or discrete controls, the direction of the knob or lever should be as shown in Figure 7.4.5.

Figure 7.4.5

Continuous control movement.



1. The maximum force, speed, accuracy, or range of the control should not exceed the limits of the least capable operator.
2. The normal control operation should be considerably less than the maximum capability of most operators.
3. The number of controls should be minimized.
4. Control movement should be simple and easy to perform.
5. Natural control movements are more efficient and less fatiguing.
6. Control movement should be as short as possible, consistent with the requirements of accuracy and feel.
7. Controls should have sufficient resistance to reduce the possibility of inadvertent activation.
8. Where a single application of force or short continuous force is required, the maximum resistance should be one-half the operator's maximum strength.

9. For continuously operated controls over long periods, the resistance should be low.
10. Where power assistance is required, artificial resistance cues should be provided.
11. Controls should be designed to withstand abuse.
12. Controls should provide a positive indication of activation so that malfunctions can be identified.
13. Controls should indicate a positive indication of system response.
14. The control surfaces should be designed to prevent slippage of the activating human appendage.
15. Hand controls are preferred in the following situations:

    - if the accuracy of the control is important
    - if the speed on control positioning is important
    - if continuous or prolonged force (greater than 20 pounds) is not necessary

16. The use of foot controls is recommended in the following cases:

    - when continuous, nonprecise control is required
    - where moderate to large forces are required (greater than 20 pounds)
    - when the hands are likely to become overburdened

Physical Requirements--Material Handling

One of the most common physical tasks performed by the worker on the job is the handling of material. Many injuries result from manual material handling by the worker. Improper handling of material often causes strains, sprains, fractures, and bruises.

There are many causes of improper lifting and carrying materials. Often the worker attempts to carry material that is too heavy. In some cases though the object may not be too heavy, it may be too bulky to maintain an adequate grip. The surface of the material may be slippery, thus causing the worker tolose grip during carrying. Rough or jagged surfaces of the material being handled can cause minor cuts and scratches which, when they occur, may cause the worker to drop the material, resulting in even a greater potential for serious injury. Finally, in some cases the worker must handle hazardous material, either manually or through the use of a system in which the worker must be in close proximity.

A major function of the industrial hygiene engineer who designs a job is controlling its material handling hazards. One of the questions that should be asked is, "Can the job be changed to eliminate the need to lift or carry the materials?" In some cases a gravity feed mechanism may be provided to move the materials into the machine from their locations. Mechanical or pneumatic conveyors can be used to move the material, either on a continuous or intermittent basis. One obvious method for material handling is the use of forklift trucks and/or hand trucks to move the material. The placement of the materials in a position for easy machine feeding is important. The materials should be placed at the same level as they are fed into the machine in order to remove the necessity for the operator to bend down and pick up the materials to be fed to the machine. Finally, bulk feeding of the materials in their packaged format may be possible.

The industrial hygiene engineer should consider if the objects being handled are hazardous to the worker. If these objects are hazardous and cannot be substituted for by other materials, then it is necessary to protect the worker from the hazardous materials. Automatic handling with the worker in a remote location is certainly a possibility. However, it may be necessary for the worker at times to come into physical contact with the materials. In such cases, personal protective equipment, such as gloves, aprons, respirators, etc., may be necessary to provide personal protection to the worker.

In some special types of material handling it may be necessary to design methods to provide assistance for handling the materials. Simple aids, such as tote boxes with handles, may be appropriate. Carts, pulleys, and hoists are also potential methods for assisting in the material handling process. If manual material handling is still necessary, it may be necessary that additional crew members be available to handle the material.

Finally, the employees involved in material handling should be trained to recognize potential material handling hazards. In this manner, the employeeswill become aware of the need to use proper techniques. The employee should be trained in the proper lifting and handling techniques to use for the various kinds of materials that are used on the job.

Summary

One of the major ways in which the field of ergonomics can be applied to the control of potential hazard exposures is in the design of the job itself. Such a job design should consider the various functions that the individual worker must perform while on the job. These functions will include data gathering, data processing, controlling of the system, and the various physical tasks required of the system. Each of these functions must be analyzed to determine how improvements can be made to protect the worker. Though not discussed in this chapter, a potential method for reducing ergonomic hazards is the proper design of hand tools used by the worker. However, this approach is beyond the scope of this discussion. Various texts on biomechanics and machine guarding have been written that cover this subject well. It is recommended that the industrial hygiene engineer consult these texts before completing a job design to eliminate potential hazards.